

Praxis Game Theory Notebook Stream Outline

This would be a continuation and re-framing of concepts taught in ECON 221 and 421. Classical game theory assumed the players were rational humans computing equilibria. Increasingly the players are **algorithms learning against each other** (pricing bots, auction bidders, recommender systems, LLM agents, etc). Instead of asking to find an equilibrium we will look at what these algorithms converge to.

Pitch can be: "You learned to solve games by hand in 221/421 (Nash, SPE, Mixed Nash) But increasingly the players aren't people reasoning to an equilibrium; they're algorithms learning against each other: pricing bots, auction bidders, recommender systems, LLM agents. So instead of only looking for equilibria we look at what learning agents converge to."

- This is how we can hit both the core game theory concepts we would want to teach (221,421), but also have the ML/AI framing to fit the project.

Outline

Notebook 1: Intro + simpler best response (Econ 221 level)

- 2x2 game and a single decision, how do we react to an opponent's move.
- Dominance
- Best responses.
- We build a basic dynamic best-response simulator (it reacts to the opponents last move only).
- Coordination games are solved, but matching pennies breaks this.

Goal: encode a game, compute best responses + pure NE in code, explain why BR-dynamics converge in some games and cycle in others

Notebook 2: Mixed Strategies and learning to predicting the opponent (~ Econ 421 Level)

- Instead of just one move the agent reacts to a player's whole history, it forms a belief (beliefs and mixed strategy).
- Mixed Strategies come out of the agent's learning.
- Build an ML "player": predict opponent (belief) and then can best-respond; show convergence plots on Matching Pennies

- Mirroring NB1 we then show Shapley's game where mixed does not converge and cycles.

Goal: implement a belief-based learner, show frequencies convergence to a mixed NE, explain the belief interpretation of mixing and when it fails

Notebook 3: Repeated Games and Cooperation (Econ 421+)

- Introduce repeated games, and look at concepts of one-shot defection or long term cooperation
- Iterated Prisoners Dilemma and rules over history (tit-for-tat, grudger/grim-trigger, win-stay-lose-shift, always-defect)
- Run an Axelrod-style tournament: pit strategies against each other (Python Axelrod Library), show tit-for-tat's success
- Link all this to folk theorem (in infinitely repeated games, virtually any long-run outcome can be sustained as an equilibrium)

Goal: run a repeated-game tournament, explain how repetition sustains cooperation that the one-shot Nash rules out, describe the common strategies (tit-for-tat, grim-trigger) and why retaliation/forgiveness matter, and connect this to the folk theorem.

Notebook 4: Learning Agents and Collusion (Masters)

- Put two independent agents (who are using Q-learning) into a repeated Bertrand pricing game. They can't communicate and they learn supracompetitive prices (smaller replication of <https://www.aeaweb.org/articles?id=10.1257/aer.20190623>).
- We plot the price climbing above the Nash equilibrium
- We also look at one-shot Nash and how the agents learned punishment strategies
- We look at this from an anti-trust perspective of what can go wrong when pricing is left in the hands of algorithms/ML.

Goal: train independent Q-learners, show prices in infinite horizon > one-shot Nash in Bertrand pricing. Discuss the anti-trust perspective.

Sources:

- <https://www.aeaweb.org/articles?id=10.1257/aer.20190623>
- https://lihao.microeconomics.ca/li_content/econ221/outline.html (221 syllabus)
- https://lihao.microeconomics.ca/li_content/econ421/outline.html (421 syllabus)
- <https://nashpy.readthedocs.io>
- <https://axelrod.readthedocs.io/en/stable/>
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